

# RITHM: Recursive Information-Theoretic Halting for Multimodal Sentiment Understanding

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## Abstract

Multimodal affective understanding predicts continuous, ordinal, or categorical affect from language, acoustic, and visual cues, but cross-modal reliability and reasoning difficulty vary substantially across samples. Previous fusion methods often collapse all modalities into a fixed representation or allocate the same computation depth to every utterance, which may amplify noisy auxiliary cues for simple cases while under-refining ambiguous or conflicting ones. Therefore, we propose **RITHM** (Recursive Information-Theoretic Halting Model), a text-anchored framework for sample-adaptive multimodal reasoning: (i) cascaded variational information bottlenecks distill task-relevant acoustic and visual evidence into compact latent representations; (ii) recursive latent refinement progressively updates the bottleneck state according to sample difficulty rather than assuming a fixed reasoning depth; and (iii) information-theoretic halting and syntax-aware token composition jointly control adaptive computation and inject hierarchical linguistic structure. We evaluate RITHM on MSA and MHD tasks, showing that adaptive bottleneck refinement provides a unified mechanism for robust multimodal affective understanding across regression and classification settings.

## 1 Introduction

Multimodal affective computing aims to infer human opinions and affective states from language, acoustic cues, and visual behavior. It covers a broad set of tasks, including multimodal sentiment analysis (MSA) and multimodal humor detection (MHD). In these tasks, language often provides dense semantic content, while vocal delivery and facial expressions offer complementary signals that can confirm, weaken, or reverse the literal meaning of an utterance (Zadeh et al., 2016, 2018; Hasan et al., 2019; Castro et al., 2019). The central challenge is therefore not simply to combine modalities,

but to decide which cross-modal evidence should influence the prediction for a particular sample.

Recent MSA methods have made substantial progress by designing stronger fusion operators, including tensor fusion, cross-modal Transformers, modality-disentanglement, and information bottleneck architectures (Zadeh et al., 2017; Tsai et al., 2019; Hazarika et al., 2020; Mai et al., 2022; Xiao et al., 2024). These models differ in how they exchange information across modalities, but they often use a fixed computation path for all samples. This is restrictive because multimodal utterances vary in how much cross-modal reasoning they require: some are resolved by clear textual evidence, whereas others depend on subtle conflicts between literal content, prosody, and facial behavior. Uniform fusion depth may therefore either introduce unnecessary auxiliary-modal noise or stop before difficult samples have been sufficiently refined.

We propose **RITHM** (Recursive Information-Theoretic Halting Model), a text-anchored framework that makes multimodal reasoning both *structure-aware* and *sample-adaptive*. RITHM first uses the language representation as the primary semantic state and progressively distills acoustic and visual evidence through cascaded variational bottlenecks. It then recursively refines the latent state and monitors the information gain between successive states. When the gain saturates, the model halts; otherwise, it continues refinement up to a bounded depth. In parallel, a syntax-aware token composer, supervised by silver constituency spans, builds hierarchical textual representations for sentiment-bearing structures.

We first demonstrate the effectiveness of RITHM in the context of MSA. Afterwards, we extend RITHM to classification tasks (MHD) (Hasan et al., 2019) to verify the generalizability of RITHM.

RITHM’s contributions are:

- **Adaptive multimodal refinement.** We in-

introduce a recursive information-theoretic bottleneck that progressively filters acoustic and visual evidence under a text-guided pathway.

- **Structure-aware halting.** We combine information-gain-based halting with syntax-aware token composition, allowing samples to use different refinement depths while preserving a stronger linguistic anchor.
- **Controlled empirical validation.** Experiments on CMU-MOSI, CMU-MOSEI, and UR-FUNNY v2, together with ablation and fixed-depth comparisons, show when adaptive refinement helps and where task-specific pragmatic modeling remains useful.

## 2 Related Work

**Multimodal Sentiment Analysis.** MSA methods have evolved from feature concatenation to interaction-based architectures. Tensor and low-rank fusion methods explicitly model high-order cross-modal interactions (Zadeh et al., 2017; Liu et al., 2018), while cross-modal attention methods such as MulT learn directed dependencies among unaligned language, acoustic, and visual sequences (Tsai et al., 2019). Another line of work improves multimodal representations by separating modality-invariant and modality-specific factors or by introducing modality-aware self-supervision (Hazarika et al., 2020; Yu et al., 2021). Recent studies further address robustness through causality-aware fusion, knowledge-guided dynamic weighting, language-focused disentanglement, and textual descriptions of audio-visual emotion cues (Huang et al., 2025a; Feng et al., 2024; Wang et al., 2025; Wu et al., 2025b). These methods strengthen the quality of fused representations, but most still execute a fixed computation graph for every sample. RITHM differs by allocating recursive computation according to the latent information gain of each utterance.

**Information Bottleneck in Multimodal Learning.** The Information Bottleneck (IB) principle learns compact representations that preserve task-relevant information while suppressing nuisance variation. Multimodal IB methods use this principle to reduce redundant or unreliable modality information (Mai et al., 2022, 2023). Extensions have studied consistency-specificity separation, theoretically constrained optimal MIB objectives, causal sufficiency-necessity balancing under cross-modal

uncertainty, and double-bottleneck designs for robust MSA (Liu et al., 2024; Wu et al., 2025a; Chen et al., 2025; Huang et al., 2025b; Xie et al., 2024). ITHP further designates a primary modality and treats the remaining modalities as detectors in an information pathway, producing compact latent states for multimodal learning (Xiao et al., 2024). In general, these methods compress a representation at a fixed stage of the network. RITHM embeds bottleneck regularization inside a recursive refinement loop, allowing compression and reasoning depth to adapt jointly to the input.

**Adaptive Computation and Syntax-Aware Modeling.** Adaptive computation time dynamically allocates inference steps according to input difficulty (Graves, 2016) and has been extended to variable-depth neural architectures (Dehghani et al., 2018). In parallel, syntax-aware representation learning introduces hierarchical linguistic structure through recursive composition and constituency-guided encoding (Socher et al., 2013; Kitaev and Klein, 2018). These two research directions address complementary questions: adaptive computation decides how long a model should reason, while syntax-aware modeling improves what the text representation contains. Existing multimodal fusion methods rarely combine these ideas. RITHM unifies syntax-guided recursive composition, cascaded variational bottlenecks, and information-theoretic halting within a single multimodal sentiment framework.

## 3 Methodology

### 3.1 Overview

Given an utterance with text, acoustic, and visual modalities, we denote the input as

$$X = (X_T, X_A, X_V), \quad (1)$$

where  $X_T$  is the word sequence, and  $X_A$  and  $X_V$  are the aligned acoustic and visual feature sequences, respectively. The goal is to predict either a continuous sentiment score or a binary pragmatic label such as humor.

Directly concatenating modalities can amplify modality dominance and unaligned noise. From an information-theoretic perspective (Tishby et al., 2000; Xiao et al., 2024), multimodal learning should compress the primary modal state while retaining task-relevant information from auxiliary modalities. A fixed fusion path cannot adjust this

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**Algorithm 1** RITHM forward and training overview
 

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**Require:** Text  $X_T$ , acoustic features  $X_A$ , visual features  $X_V$ , halting threshold  $\tau$ , maximum depth  $L$

**Ensure:** Prediction  $\hat{y}$

- 1: Encode text with the PLM:  $H_T, x_T \leftarrow \text{PLM}(X_T)$ .
  - 2: Sample  $b_0 \sim q(b_0|x_T)$  and reconstruct acoustic evidence  $\hat{x}_A$ .
  - 3: Sample  $b_1 \sim q(b_1|b_0)$  and reconstruct visual evidence  $\hat{x}_V$ .
  - 4: Initialize the recursive state  $b^{(0)} \leftarrow b_1$ .
  - 5: **for**  $t = 0$  to  $L - 1$  **do**
  - 6:    $\tilde{b} \leftarrow \text{LayerNorm}(b^{(t)} + r_\theta(b^{(t)}))$ .
  - 7:    $\Delta^{(t)} \leftarrow \frac{1}{d} \|\tilde{b}_{\text{cls}} - b_{\text{cls}}^{(t)}\|_1$ .
  - 8:    $b^{(t+1)} \leftarrow \tilde{b}$ .
  - 9:   **if**  $\Delta^{(t)} < \tau$  **then**
  - 10:     **break**
  - 11:   **end if**
  - 12: **end for**
  - 13: Inject the final bottleneck state into token states and compose a syntax root  $h_{\text{syntax}}$ .
  - 14: Fuse pooled multimodal evidence with  $h_{\text{syntax}}$  and predict  $\hat{y}$ .
  - 15: Optimize the task, bottleneck, and syntax losses jointly.
  - 16: **return**  $\hat{y}$
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trade-off across samples. RITHM therefore uses text as a semantic anchor and selectively extracts acoustic and visual cues that are compatible with that anchor. A pretrained language model (PLM) first encodes the text into token-level representations  $H_T$  and a pooled global representation  $x_T$ :

$$H_T, x_T = \text{PLM}(X_T). \quad (2)$$

The pooled textual state  $x_T$  initializes the multimodal bottleneck path, while the token-level states  $H_T$  are later enhanced by the refined bottleneck state before syntax-aware composition. The framework contains three components: a text-guided variational bottleneck for cross-modal filtering, recursive latent refinement with adaptive halting for dynamic computation, and syntax-aware token composition for hierarchical linguistic structure. Algorithm 1 summarizes the overall forward and training flow. A compact notation summary and additional training details are provided in Appendix C.

### 3.2 Text-Guided Variational Bottleneck

The first component compresses multimodal evidence through stochastic latent variables. Standard early fusion, such as feature concatenation, can mix useful signals with modality-specific noise. RITHM instead uses a cascaded variational bottleneck in which the textual state sequentially queries acoustic and visual features and preserves cues relevant to the semantic context.

First, the textual representation  $x_T$  is mapped into a Gaussian posterior that captures the text-audio dependency:

$$\mu_0, \log \sigma_0^2 = f_0(x_T), \quad (3)$$

from which a latent state is sampled using the reparameterization trick:

$$b_0 = \mu_0 + \epsilon \odot \sigma_0, \quad \epsilon \sim \mathcal{N}(0, I). \quad (4)$$

An acoustic decoder reconstructs the pooled acoustic feature  $x_A$  from  $b_0$ :

$$\hat{x}_A = g_0(b_0). \quad (5)$$

The second bottleneck maps the acoustic-aware state  $b_0$  into a visual-aware latent state  $b_1$ :

$$\mu_1, \log \sigma_1^2 = f_1(b_0), \quad b_1 = \mu_1 + \epsilon \odot \sigma_1, \quad (6)$$

and reconstructs the pooled visual representation  $x_V$ :

$$\hat{x}_V = g_1(b_1). \quad (7)$$

Unlike parallel cross-attention mechanisms, the cascaded bottleneck progressively constrains information flow across modalities. The final latent state  $b_1$  is therefore encouraged to preserve predictive cross-modal evidence while suppressing modality-specific variation.

### 3.3 Recursive Refinement and Adaptive Halting

The second component reasons over the visual-aware state  $b_1$ . Fixed-depth networks can under-process ambiguous utterances and over-smooth direct expressions. RITHM instead applies a residual recursive encoder with an adaptive halting rule.

Starting with  $b^{(0)} = b_1$ , the state is recursively refined:

$$b^{(t+1)} = \text{LayerNorm}(b^{(t)} + r_\theta(b^{(t)})). \quad (8)$$

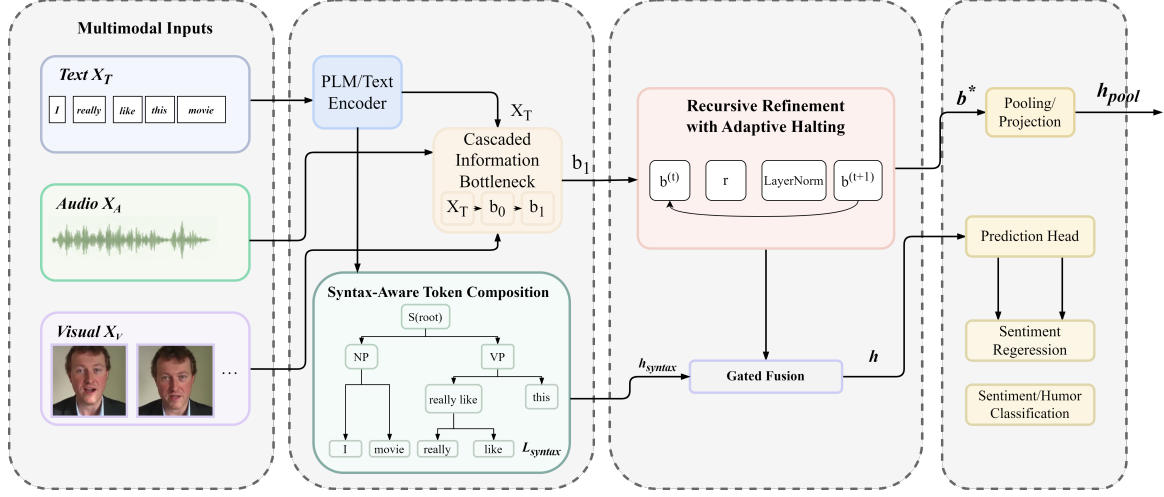


Figure 1: Overview of RITHM. The framework first constructs text-anchored multimodal evidence through cascaded bottlenecks, then performs recursive latent refinement with adaptive halting, and finally applies syntax-aware composition before downstream prediction.

Rather than applying a static number of refinement steps, RITHM monitors the magnitude of state updates at each iteration:

$$\Delta^{(t)} = \frac{1}{d} \left\| b_{\text{cls}}^{(t+1)} - b_{\text{cls}}^{(t)} \right\|_1, \quad (9)$$

where  $b_{\text{cls}}^{(t)}$  denotes the normalized CLS-position bottleneck state and  $d$  is the hidden dimension. If  $\Delta^{(t)} < \tau$ , the representation is treated as converged and recursion halts. Otherwise, refinement continues up to maximum depth  $L$ . This rule gives each sample a different computation budget: straightforward utterances tend to converge early, whereas ambiguous or cross-modally conflicting examples receive additional refinement before prediction.

### 3.4 Syntax-Aware Token Composition

Sentiment and pragmatics are compositional. Contrast, negation, and clause-level modification can change the polarity of local expressions. To model this structure, the third component introduces a recursive token composition module over multimodal-enhanced token states  $\tilde{H}_T$ , obtained by injecting the refined bottleneck state back into the PLM token representations.

At each step, the module considers every adjacent pair of current spans  $(s_i, s_{i+1})$ . Let  $u_i$  and  $u_{i+1}$  denote their current vector states. A candidate parent representation is first computed by a shared composition network,

$$c_i = \text{LayerNorm} \left( f_c([u_i; u_{i+1}] + \frac{1}{2}(u_i + u_{i+1})) \right), \quad (10)$$

where  $[\cdot; \cdot]$  denotes concatenation. The merge score for this candidate is then produced by a learned scalar scorer:

$$a_i = f_m([u_i; u_{i+1}; c_i]), \quad (11)$$

$$p_i^{\text{merge}} = \text{softmax}(a_i/T), \quad (12)$$

where  $T$  is the syntax temperature. During the forward composition pass, the highest-scoring adjacent pair is selected, replaced by its composed state  $c_i$ , and the process repeats until one sentence-level syntax root remains.

When silver constituency-span caches are available, they supervise the merge distribution at each reduction step. The target distribution  $q^{\text{silver}}$  places mass on adjacent merges whose combined span boundary appears in the silver constituency cache; if no candidate at a step matches a cached span, that step is left unsupervised. The supervised merge loss is:

$$\mathcal{L}_{\text{syntax}} = - \sum_i q_i^{\text{silver}} \log p_i^{\text{merge}}. \quad (13)$$

The resulting syntax-aware root representation,  $h_{\text{syntax}}$ , is fused with the multimodal bottleneck output, denoted as  $h_{\text{pool}}$ , through a learned gate:

$$h = \text{LayerNorm}(h_{\text{pool}} + \zeta \odot h_{\text{syntax}}). \quad (14)$$

This composition encourages predictions to use structural boundaries rather than only local token correlations. It is especially useful for sentiment-bearing expressions involving contrast, negation, or delayed polarity shifts.

### 3.5 Training Objective

Training combines task supervision with reconstruction-regularized bottleneck terms. Following variational information bottleneck practice (Alemi et al., 2016; Xiao et al., 2024), we upper-bound the compression term with KL divergence to a standard Gaussian prior and approximate the modality-retention term with reconstruction losses under fixed-variance Gaussian decoders:

$$\mathcal{L}_{IB}^{(0)} = D_{KL}(q(b_0|x_T)||p(b_0)) + \beta \cdot \text{MSE}(\hat{x}_A, x_A), \quad (15)$$

$$\mathcal{L}_{IB}^{(1)} = D_{KL}(q(b_1|b_0)||p(b_1)) + \gamma \cdot \text{MSE}(\hat{x}_V, x_V). \quad (16)$$

For example, minimizing  $\text{MSE}(\hat{x}_A, x_A)$  corresponds to maximizing a variational reconstruction term  $\mathbb{E}[\log q_\psi(x_A|b_0)]$  up to a constant when  $q_\psi$  is a fixed-variance Gaussian likelihood. The visual reconstruction term is defined analogously.

The overall information bottleneck loss is:

$$\mathcal{L}_{IB} = \mathcal{L}_{IB}^{(0)} + \lambda \mathcal{L}_{IB}^{(1)}. \quad (17)$$

For continuous sentiment regression, the total loss combines the primary regression objective ( $\mathcal{L}_{reg}$ ), classification objective, syntax supervision, and the bottleneck penalty:

$$\mathcal{L} = \mathcal{L}_{reg} + \eta_s \mathcal{L}_{syntax} + \eta_{IB} \mathcal{L}_{IB}. \quad (18)$$

For binary classification tasks such as humor detection,  $\mathcal{L}_{reg}$  is replaced with binary cross-entropy.

## 4 Experiments

We evaluate RITHM on two continuous sentiment benchmarks and one pragmatic classification benchmark. The experiments are designed to answer three questions: (1) whether RITHM improves sentiment intensity prediction, (2) which architectural components account for the gains, and (3) whether adaptive halting behaves consistently with multimodal reasoning difficulty.

### 4.1 Datasets and Evaluation Metrics

#### 4.1.1 MSA

We evaluate multimodal sentiment analysis on CMU-MOSI and CMU-MOSEI, two widely used benchmarks containing aligned language, acoustic,

and visual signals with real-valued sentiment annotations. CMU-MOSI contains 2,199 utterances, split into 1,284 training, 229 validation, and 686 test samples; CMU-MOSEI contains 22,856 utterances, split into 16,326 training, 1,871 validation, and 4,659 test samples. We report MAE and Pearson correlation to evaluate continuous sentiment intensity, and Acc-7, Acc-2, and F1 to evaluate thresholded sentiment classification. Following standard practice, Acc-2 and F1 are computed after excluding zero-valued neutral instances. Metric definitions and dataset statistics are provided in Appendix D and Table 3.

#### 4.1.2 MHD

We also evaluate multimodal humor detection on UR-FUNNY v2 as a pragmatic transfer diagnostic. UR-FUNNY v2 is a binary classification benchmark built from multimodal punchline contexts, with 7,614 training, 980 validation, and 994 test samples. We report Acc-2 and F1 for RITHM under the matched split used by our implementation. Because published humor-detection baselines differ in feature pipelines, backbone choices, and whether F1 is reported, we use this setting as a compact diagnostic for syntax-aware recursive refinement rather than as a fully protocol-aligned leaderboard.

## 4.2 Implementation Details

### 4.2.1 Backbones

For CMU-MOSI and CMU-MOSEI, RITHM uses DeBERTa-v3 as the textual encoder and combines it with the acoustic and visual features provided by the dataset pipeline. For UR-FUNNY v2, we use an ALBERT-base text encoder to match the HKT-style binary setup used by the comparison protocol. The same feature pipeline, train/dev/test splits, and silver syntax-span caches are used across the full model and all controlled variants. A compact summary of backbone and model hyperparameters is provided in Appendix B.

### 4.2.2 Optimization

For the sentiment benchmarks, hyperparameters are selected with Optuna using validation MAE, and checkpoints are selected by validation MAE before test evaluation. For UR-FUNNY v2, the binary model is selected by validation accuracy. Unless otherwise stated, all ablations and sensitivity analyses keep the selected configuration fixed and change only the component under study. The final settings use train batch sizes of 8, 8, and 32

for MOSI, MOSEI, and UR-FUNNY v2, learning rates of  $1 \times 10^{-5}$  for all datasets, and maximum recursion depths of 5, 3, and 3, respectively. The selected halting thresholds are 0.04, 0.02, and 0.06, respectively. Full hyperparameter values are listed in Table 4.

### 4.3 Main Results: Multimodal Sentiment

Table 1 reports results on CMU-MOSI and CMU-MOSEI.

**Analysis.** Table 1 shows that RITHM is particularly effective on metrics that measure sentiment intensity. On CMU-MOSI, it obtains the best results in this comparison on Acc-7, Acc-2, F1, MAE, and Corr. On CMU-MOSEI, it achieves the best MAE (0.498) and Corr (0.801), while the strongest published baselines remain ahead on Acc-7, Acc-2, and F1. We therefore frame the main empirical claim as improved continuous sentiment estimation rather than uniform dominance over every classification-oriented metric.

The gains are more pronounced on regression metrics than on thresholded classification metrics, especially on CMU-MOSEI. This pattern is consistent with the training objective: RITHM selects checkpoints for continuous sentiment estimation, whereas Acc-2 and F1 are obtained by thresholding real-valued predictions into polarity labels. Models optimized more directly for discrete separation can remain stronger near the decision boundary, while RITHM is better calibrated for graded sentiment intensity. This supports the intended role of the text-guided bottleneck and recursive refinement mechanism: they preserve task-relevant cross-modal evidence while suppressing modality-specific noise, leading to more stable continuous prediction.

### 4.4 Transfer Diagnostic: Humor Detection

We further evaluate RITHM on UR-FUNNY v2 to test whether syntax-aware recursive refinement transfers beyond continuous sentiment regression. Published UR-FUNNY v2 results follow heterogeneous reporting conventions and often omit F1. The source comparison tables for most baselines report accuracy and parameter counts but not class-level predictions or F1, so we cannot recompute comparable F1 values for those methods. We therefore use a compact MHD-style comparison in the main text rather than treating this setting as a complete humor-detection leaderboard. As shown in Figure 2, RITHM obtains 75.35% Acc-2 and 75.52 F1. Because most baselines do not report F1, the

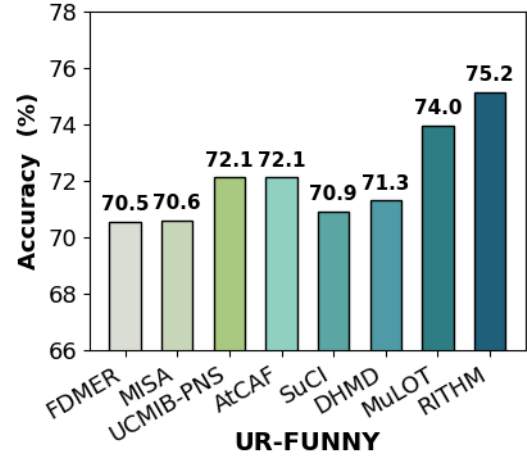


Figure 2: Accuracy on UR-FUNNY v2(Hasan et al., 2019).

75.52 value should be interpreted as an internal reference for RITHM under our matched split rather than as a ranked cross-model comparison. The result suggests that adaptive bottleneck refinement can transfer to pragmatic inference, while specialized humor objectives may still be beneficial.

### 4.5 Ablation Study

To analyze the contribution of each component, we perform ablations on CMU-MOSI by modifying one mechanism at a time while keeping all other settings unchanged. All variants use the same DeBERTa-v3-base backbone, feature pipeline, and Optuna-tuned hyperparameters. These controlled variants isolate the contribution of each mechanism rather than serving as externally sourced baselines. We also include a *Fixed-Depth* setting ( $L = 5$ ) to test whether the improvement comes from adaptive halting rather than merely from adding more refinement steps.

**Effect of Syntax Supervision.** For the syntax-aware component, we remove the explicit syntax supervision by setting  $\eta_s = 0$  while retaining the syntax composition pathway. This variant increases MAE from 0.580 to 0.617 (+0.037) and reduces F1 from 89.5 to 87.2. Because the architecture still contains the syntax pathway, the degradation mainly reflects the loss of span-level structural guidance rather than the removal of extra parameters. This indicates that syntax supervision helps RITHM form a stronger textual anchor for contrast, negation, and compositional sentiment cues.

**Effect of Recursive Refinement.** For the recursive

Table 1: Comparison results on CMU-MOSI and CMU-MOSEI. Acc-2 and F1 are computed under the non-zero protocol, where samples with zero sentiment scores are excluded. Results marked with  $\dagger$  are taken from the original papers, while the others are reproduced in our experiments. The best and second-best results are highlighted in bold and underlined, respectively (Zadeh et al., 2016, 2018).

| Model                                  | CMU-MOSI        |                 |               |                  |                 | CMU-MOSEI       |                 |               |                  |                 |
|----------------------------------------|-----------------|-----------------|---------------|------------------|-----------------|-----------------|-----------------|---------------|------------------|-----------------|
|                                        | Acc7 $\uparrow$ | Acc2 $\uparrow$ | F1 $\uparrow$ | MAE $\downarrow$ | Corr $\uparrow$ | Acc7 $\uparrow$ | Acc2 $\uparrow$ | F1 $\uparrow$ | MAE $\downarrow$ | Corr $\uparrow$ |
| MFM (Tsai et al., 2018)                | 33.3            | 80.0            | 80.1          | 0.948            | 0.664           | 50.8            | 83.4            | 83.4          | 0.580            | 0.722           |
| Self-MM (Yu et al., 2021)              | 45.8            | 84.9            | 84.8          | 0.731            | 0.785           | 53.0            | 85.2            | 85.2          | 0.540            | 0.763           |
| DLF $\dagger$ (Wang et al., 2025)      | 47.1            | 85.1            | 85.0          | 0.731            | 0.781           | 53.9            | 85.4            | 85.3          | 0.536            | 0.764           |
| KuDA $\dagger$ (Feng et al., 2024)     | 47.1            | 86.4            | 86.5          | 0.705            | 0.795           | 52.9            | 86.5            | 86.6          | 0.529            | 0.776           |
| DEVA $\dagger$ (Wu et al., 2025b)      | 46.3            | 86.3            | 86.3          | 0.730            | 0.787           | 52.3            | 86.1            | 86.2          | 0.541            | 0.769           |
| C-MIB (Mai et al., 2022)               | 47.7            | 87.8            | 87.8          | 0.662            | 0.835           | 52.7            | 86.9            | 86.8          | 0.542            | 0.784           |
| ITHP (Xiao et al., 2024)               | 47.7            | 88.5            | 88.5          | 0.663            | <u>0.856</u>    | 52.2            | <b>87.1</b>     | <b>87.1</b>   | 0.550            | 0.792           |
| Multimodal Boosting (Mai et al., 2023) | <u>49.1</u>     | 88.5            | 88.4          | 0.634            | 0.855           | 54.0            | 86.5            | 86.5          | 0.523            | 0.779           |
| HME (Zhuang et al., 2026)              | 48.3            | 88.1            | 88.1          | <u>0.620</u>     | <u>0.856</u>    | <u>54.3</u>     | <u>86.8</u>     | 86.9          | <u>0.513</u>     | 0.792           |
| <b>RITHM (Ours)</b>                    | <b>52.0</b>     | <b>89.5</b>     | <b>89.5</b>   | <b>0.580</b>     | <b>0.864</b>    | <b>54.8</b>     | <b>87.1</b>     | <u>87.0</u>   | <b>0.498</b>     | <b>0.801</b>    |

Table 2: Ablation and fixed-depth comparison on CMU-MOSI. Lower MAE and higher F1 are better;  $\Delta$ MAE is relative to Full RITHM.

| Config.           | MAE $\downarrow$ | $\Delta$ | F1 $\uparrow$ | Steps       |
|-------------------|------------------|----------|---------------|-------------|
| <b>Full RITHM</b> | <b>0.580</b>     | –        | <b>89.5</b>   | <b>3.18</b> |
| w/o Syntax Sup.   | 0.617            | +0.037   | 87.2          | 3.25        |
| w/o Recursion     | 0.621            | +0.041   | 86.9          | 1.00        |
| Fixed $L = 5$     | 0.595            | +0.015   | 88.4          | 5.00        |

refinement component, we remove iterative latent updates by setting the recursion depth to one, so the model predicts from a single-pass bottleneck state. This produces the largest degradation in Table 2, increasing MAE from 0.580 to 0.621 (+0.041) and reducing F1 from 89.5 to 86.9. The result shows that recursive refinement is not merely an architectural embellishment; it provides the main mechanism for progressively denoising and reconciling multimodal evidence before prediction.

**Effect of Adaptive Halting.** For adaptive halting, we keep recursive refinement but disable sample-dependent stopping by forcing every example to execute the maximum depth  $L = 5$ . Although this fixed-depth variant is better than removing recursion entirely, it remains worse than full RITHM: MAE increases from 0.580 to 0.595 (+0.015), F1 drops from 89.5 to 88.4, and the average number of refinement steps rises from 3.18 to 5.00. This confirms that the benefit of RITHM comes not only from having more refinement capacity, but also from allocating that capacity selectively according to each sample’s convergence behavior.

## 4.6 Sensitivity Analysis

We further examine how RITHM responds to the two hyperparameters that directly control recursive computation: the halting threshold  $\tau$  and the maximum recursion depth  $L$ . The analysis is conducted on CMU-MOSI while keeping the remaining hyperparameters fixed to the selected configuration. Figure 3 reports both Acc-7 and MAE, with the MAE axis inverted so that upward movement consistently indicates better performance.

The model is most stable around the selected threshold  $\tau = 0.04$ , where it obtains the best MAE (0.580) and Acc-7 (51.97%). Smaller thresholds such as 0.015 and 0.02 make the halting rule too conservative: almost all samples reach the maximum depth, increasing average recursion to 5.0 steps and worsening MAE to 0.615. Larger thresholds halt too early, reducing the average depth to roughly 2.0 steps at  $\tau \geq 0.06$  and again degrading both MAE and Acc-7. This pattern shows that RITHM benefits from an intermediate halting criterion rather than from always refining to the maximum depth or exiting as soon as possible.

Varying  $L$  leads to a consistent conclusion. A single refinement step under-processes the representation, while shallow limits such as  $L = 2$  restrict the adaptive policy and produce the weakest performance. Increasing the available budget improves the best result at  $L = 5$ , but the observed average depth remains only 3.18 steps. Thus, the gain does not come from forcing every sample through all refinement layers; it comes from giving the model enough capacity to spend additional computation only when the latent state has not converged.

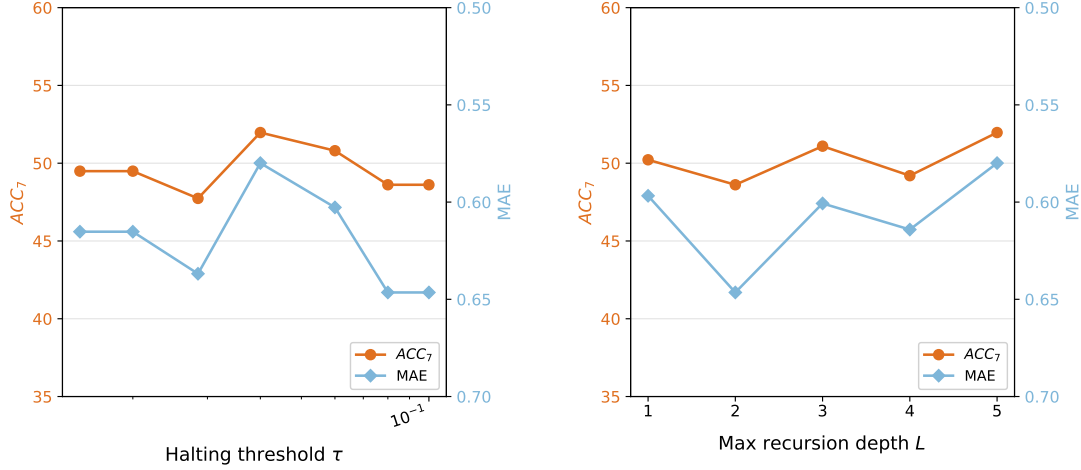


Figure 3: Sensitivity of RITHM to the halting threshold  $\tau$  and maximum recursion depth  $L$  on CMU-MOSI. Acc-7 is shown on the left axis, and MAE is shown on the inverted right axis so that higher curves indicate better performance.

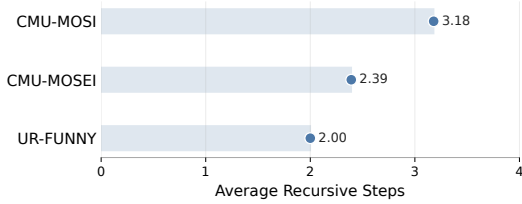


Figure 4: Average recursive halting steps across datasets.

#### 4.7 In-depth Analysis

Figure 4 reports the average number of active latent refinement states on the test split. RITHM uses the deepest average computation on MOSI (3.18), fewer steps on MOSEI (2.39), and the shallowest computation on UR-FUNNY v2 (2.00). The pattern indicates that the halting mechanism does not apply a uniform reasoning budget; instead, it changes the amount of latent refinement according to the stability of each dataset’s learned multimodal representation.

The reported recursive depth is a latent-state statistic rather than a surface property of the input. It does not count tokens, modalities, or levels in the syntax composition tree. For each example, the counter starts from the initial visual-aware bottleneck state  $b^{(0)}$  and increases only while the sample remains active in the refinement loop. Halting is triggered when the change of the normalized CLS state falls below the threshold, so a low depth means that the fused bottleneck representation has stabilized early, not that the sentence is short or syntactically shallow.

The dataset ordering should therefore be interpreted as a behavior of the trained model, not as an absolute ranking of dataset difficulty. MOSI contains short spontaneous opinion clips where sentiment can depend on local negation, prosody, and visual affect, so more examples require additional reconciliation inside the bottleneck state. MOSEI receives an intermediate average depth under the fixed test run, while UR-FUNNY v2 often provides more direct pragmatic cues in this setup and reaches stable latent states with fewer refinement steps.

This analysis supports the role of adaptive computation in RITHM. The model does not merely add a deeper recurrent block to every sample; it uses the halting criterion to stop when the multimodal bottleneck has converged and to continue when the latent state remains unstable. As a result, recursive refinement serves as a sample-dependent mechanism for resolving multimodal evidence before prediction.

## 5 Conclusion

We introduced RITHM, a text-anchored multimodal framework with cascaded bottlenecks, adaptive recursive refinement, and syntax-aware composition. Experiments show that RITHM improves continuous affect prediction while allocating computation according to sample-level uncertainty.

## Limitations

RITHM introduces additional computation from recursive latent refinement and from building the silver constituency-span cache used to supervise syntax-aware token composition. Although adaptive halting can reduce unnecessary refinement for easy samples, training still requires tuning bottleneck and halting hyperparameters, and the syntax component depends on the quality of the external parser that produces the silver spans.

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## A Dataset Statistics

Table 3 lists the split sizes used in our experiments. We keep the official train/validation/test partitions for the sentiment datasets and use the matched split adopted by our UR-FUNNY v2 reproduction, so the reported results in the main text are computed on the same evaluation partitions as the fixed configurations.

Table 3: Dataset statistics for sentiment and pragmatic benchmarks.

| Dataset     | Train  | Valid | Test  | Total  |
|-------------|--------|-------|-------|--------|
| CMU-MOSI    | 1,284  | 229   | 686   | 2,199  |
| CMU-MOSEI   | 16,326 | 1,871 | 4,659 | 22,856 |
| UR-FUNNY v2 | 7,614  | 980   | 994   | 9,588  |

The table also highlights the scale difference between the benchmarks. CMU-MOSI contains shorter, more fine-grained opinion clips, CMU-MOSEI provides a much larger sentiment corpus, and UR-FUNNY v2 evaluates whether the same architecture can transfer to a binary pragmatic inference setting.

## B Hyperparameter Configuration

Table 4 summarizes the hyperparameter settings used across all reported experiments. The two bottleneck dimensions are reported as  $B_0/B_1$ , corresponding to the acoustic-aware and visual-aware latent states in the cascaded information bottleneck. The loss weights and halting thresholds are kept fixed for the main result, ablation, and sensitivity runs unless a specific component is intentionally modified.

Table 4: Hyperparameter settings for RITHM across datasets.

| Hyperparameter         | MOSI       | MOSEI      | UR-FUNNY v2 |
|------------------------|------------|------------|-------------|
| Text Encoder           | DeBERTa-v3 | DeBERTa-v3 | ALBERT-base |
| Bottleneck dims        | 64/128     | 64/64      | 128/64      |
| Recon. weight $\beta$  | 4          | 8          | 8           |
| Recon. weight $\gamma$ | 64         | 64         | 64          |
| $\lambda$              | 0.3        | 0.3        | 0.2         |
| Halting $\tau$         | 0.04       | 0.02       | 0.06        |
| Max depth $L$          | 5          | 3          | 3           |
| Syntax $\eta_s$        | 0.1        | 0.1        | 0.1705      |
| Batch size             | 8          | 8          | 32          |
| Learning rate          | 1e-5       | 1e-5       | 1e-5        |
| Epochs                 | 20         | 10         | 10          |

The reported settings are taken from the validation-selected fixed runs: source trials 278, 95, and 29 for MOSI, MOSEI, and UR-FUNNY

v2, respectively. For MOSI and MOSEI, selection is based on validation MAE; for UR-FUNNY v2, selection is based on validation accuracy. This keeps the reported hyperparameters aligned with the fixed-experiment reproduction scripts rather than retuning them for individual analyses.

## C Notation

Table 5 collects the notation used in the method section. We separate input-level symbols, bottleneck variables, recursive-halting variables, and syntax-fusion variables to make clear that latent recursive depth is distinct from token-level syntax composition.

Table 5: Key notation used throughout the paper.

| Symbol                   | Description                             |
|--------------------------|-----------------------------------------|
| $X_T, X_A, X_V$          | Text, acoustic, visual input sequences  |
| $x_T, x_A, x_V$          | Pooled representations of each modality |
| $b_0, b_1$               | Latent bottleneck states                |
| $b^{(t)}$                | Latent state at recursive step $t$      |
| $\mu_i, \sigma_i$        | Mean and std of Gaussian posterior      |
| $\tau$                   | Adaptive halting threshold              |
| $L$                      | Maximum recursion depth                 |
| $\Delta^{(t)}$           | CLS-state update magnitude at step $t$  |
| $h_{\text{syntax}}$      | Syntax-aware root representation        |
| $\zeta$                  | Learned syntax gate                     |
| $\beta, \gamma, \lambda$ | IB loss weights                         |
| $\eta_s, \eta_{IB}$      | Syntax and IB loss coefficients         |
| $p(\cdot)$               | True data distribution                  |
| $q(\cdot)$               | Variational approximation               |

In particular,  $b^{(t)}$  denotes the latent state after recursive refinement, whereas  $h_{\text{syntax}}$  denotes the sentence-level root produced by the syntax composition module. These two quantities interact during final fusion but measure different aspects of the model: adaptive computation over the bottleneck state and hierarchical linguistic structure over token representations.

## D Evaluation Metrics

### Binary Accuracy (Acc-2).

$$\text{Acc-2} = \frac{TP + TN}{TP + TN + FP + FN} \quad (19)$$

### F1 Score.

$$F1 = 2 \cdot \frac{P \times R}{P + R}, \quad (20)$$

$$P = \frac{TP}{TP + FP}, \quad (21)$$

$$R = \frac{TP}{TP + FN}. \quad (22)$$

### Mean Absolute Error (MAE).

$$\text{MAE}(\hat{y}, y) = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad (23)$$

### Pearson Correlation (Corr).

$$\text{Corr}(x, y) = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}}. \quad (24)$$

Following standard practice (Zadeh et al., 2016), Acc-2 and F1 are computed on positive and negative samples only, excluding zero-valued neutral instances.

We do not include the preliminary CH-SIMS v2 or MUSTARD tables in the appendix because those runs use different task protocols and are weaker than specialized baselines. Reporting them without a fuller protocol discussion would distract from the paper’s main MOSI/MOSEI and UR-FUNNY v2 claims. These datasets are instead noted as future evaluation targets in the limitations.